Dan Ghincul

Timisoara, Romania | dan.ghincul01@gmail.com | linkedin.com/in/dan-ghincul | github.com/DSElement

Education

Bachelor of Science in Computer Engineering Oct. 2023 – Jul. 2027 (expected)

Universitatea Politehnica Timisoara | GPA: 9.4/10

- Relevant Coursework: Data Structures and Algorithms, Parallel and Distributed Algorithms, Operating Systems, Operating Systems Administration, Computer Architecture, Formal Languages and Compilation Techniques, Software Architecture, Networking, Databases.

Experience

AC Labs Trainee Apr. 2026 – Jul. 2026

The Access Group | C#, ASP.NET Core, EF Core, SQL Server, React, TypeScript, Ollama, Qdrant

- Built a full-stack quiz app over 10 guided labs: ASP.NET Core REST API with EF Core on SQL Server, React/TypeScript frontend, Microsoft Entra ID sign-in, server-side grading and RAG with Qdrant vector search.
- Moved its AI quiz generation from the retired GitHub Models service to a local Ollama model, driven by a Microsoft.Extensions.AI tool-calling agent that validates questions and resubmits on error.

Back-Office Workflow Automation

Independent initiative | Python, Playwright, openpyxl, PyInstaller, PowerShell

- Cut a recurring daily back-office routine from ~60–90 minutes of manual browser and spreadsheet work to ~5 minutes by scripting three Playwright workflows: report exports, spreadsheet summaries and catalog lookups.
- Made runs resilient to expired sessions, timeouts and bad input through session reuse, bounded re-login, per-record retries, CSV/XLSX validation, Decimal money handling and incremental, atomic Excel saves.
- Shipped it as a versioned, self-contained Windows release (PyInstaller, bundled Chromium, console menu, Windows-protected saved credentials, offline self-check) that runs without a Python install.

Projects

QuizMaster | *C#, ASP.NET Core, EF Core, PostgreSQL, pgvector, React, TypeScript, Gemini*

- Rebuilt my AC Labs quiz app from scratch: ASP.NET Core Identity cookie auth with admin and user roles, official, private and community quizzes with role-based permissions, and server-side grading that hides the answer key.
- Built document RAG: a background worker extracts PDF text, splits it into overlapping sentence-aware chunks and embeds them with Gemini into PostgreSQL, searched through a pgvector HNSW cosine index.
- Wrote a Gemini tool-calling agent that searches the document and submits quizzes, rejected and retried until each question has 4 distinct options, no giveaway answer and a citation to a chunk the agent actually read.

MeshDrop (2-person team) | *Dart, Flutter, Riverpod, TCP, mDNS, X25519/Ed25519*

- Co-built an AirDrop-style LAN file-sharing app: mDNS/DNS-SD discovery, a framed TCP protocol streaming 64 KB chunks with SHA-256 checksums, and ChaCha20-Poly1305 encryption keyed per transfer via X25519.
- Owned the device-trust layer: Ed25519-signed handshakes, 6-digit pairing codes derived from the handshake transcript, trust saved only after a mutually signed confirm/ack, and automatic resend of the original file once pairing succeeds.

Concurrent Stock Exchange (team lead) | *Java 25, java.util.concurrent, Maven*

- Leading the team and building most of an in-memory stock exchange: concurrent order submission, per-company order books, asynchronous matching on a thread pool via CompletableFuture, partial fills and trade history.
- Prevented deadlocks and stale trades by locking both orders in a fixed ID order and re-checking price and remaining quantity under the locks; restricted modifications to the order's owner and re-ran matching after each change.

MRET – My Reverse Engineering Tool | *Java 21, Reflection, Maven, JUnit*

- Built a CLI that turns any JAR into text or PlantUML class diagrams via reflection (members, generics, annotations, extends/implements/uses relationships), with package filters for fat JARs and renderer plugins loaded at runtime.

Parallel and Distributed Algorithms (coursework) | *C, OpenMP, MPI, Python, pandas, Matplotlib*

- Parallelized alpha-beta minimax for N×N tic-tac-toe with two OpenMP strategies and benchmarked board sizes 3–8, depths 5–11 and 2/4/8 threads, reaching a median 2.8× speedup at 8 threads.
- Simulated Raft leader election over MPI (up to 7 nodes) with randomized 150–300 ms election timeouts, term-based voting, 50 ms heartbeats, non-blocking messaging and injected leader crashes that force re-election.

Certifications

Advanced Network Security, Advanced Network Administration, Ethical Hacking 2026

Universitatea Politehnica Timisoara and T8X Industries | Cisco Packet Tracer, Kali Linux, Nmap, Metasploit

- Configured VLANs, OSPF, HSRP, NAT/PAT and ACLs and hardened switches (port security, DHCP snooping, AAA) in Packet Tracer labs; ran Nmap and Metasploit in an isolated lab, including EternalBlue against a Windows 7 VM.

Technical Skills

Languages: C#, Java, Python, C, TypeScript/JavaScript, SQL, Dart, Kotlin, x86 Assembly, Bash, PowerShell
Backend and Frontend: ASP.NET Core, ASP.NET Core Identity, EF Core, REST, React, TanStack Query, Vite, Flutter
Data and AI: PostgreSQL, pgvector, SQL Server, Qdrant, Gemini API, Ollama, Microsoft.Extensions.AI, RAG
Systems: POSIX (processes, signals, pipes, pthreads), Win32 API, OpenMP, MPI, Java concurrency, gRPC, RabbitMQ
Tools and Testing: Git, Docker Compose, Linux, Maven, Playwright, openpyxl, PyInstaller, xUnit, JUnit, Vitest
Security and Networking: VLANs, OSPF, ACLs, Cisco Packet Tracer, Kali Linux, Nmap, Metasploit, Wireshark, dnSpy
Spoken Languages: Romanian (native), English (C1, certified), Russian (B1)